239. $\frac{8\cdot 9-7\cdot 6}{5\cdot 6-9\cdot 2}$

240. $\frac{5^2-3^2}{3-5}$

241. $\frac{6^2-4^2}{4-6}$

242. $\frac{2+4(3)}{-3-2^2}$

243. $\frac{7+3(5)}{-2-3^2}$

244. $\frac{7\cdot 4-2(8-5)}{9\cdot 3-3\cdot 5}$

245. $\frac{9\cdot 7-3(12-8)}{8\cdot 7-6\cdot 6}$

246. $\frac{9(8-2)-3(15-7)}{6(7-1)-3(17-9)}$

247. $\frac{8(9-2)-4(14-9)}{7(8-3)-3(16-9)}$

Everyday Math

248. **Baking** A recipe for chocolate chip cookies calls for $2\frac{1}{4}$ cups of flour. Graciela wants to double the recipe.

1. ⓐ How much flour will Graciela need? Show your calculation. Write your result as an improper fraction and as a mixed number.
2. ⓑ Measuring cups usually come in sets with cups for $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, and 1 cup. Draw a diagram to show two different ways that Graciela could measure out the flour needed to double the recipe.

249. **Baking** A booth at the county fair sells fudge by the pound. Their award winning "Chocolate Overdose" fudge contains $2\frac{2}{3}$ cups of chocolate chips per pound.

ⓐ How many cups of chocolate chips are in a half-pound of the fudge?

ⓑ The owners of the booth make the fudge in 10-pound batches. How many chocolate chips do they need to make a 10-pound batch? Write your results as improper fractions and as a mixed numbers.

Writing Exercises

250. Explain how to find the reciprocal of a mixed number.

251. Explain how to multiply mixed numbers.

252. Randy thinks that $3\frac{1}{2}\cdot 5\frac{1}{4}$ is $15\frac{1}{8}$. Explain what is wrong with Randy's thinking.

253. Explain why $-\frac{1}{2}$, $\frac{-1}{2}$, and $\frac{1}{-2}$ are equivalent.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
multiply and divide mixed numbers.			
translate phrases to expressions with fractions.			
simplify complex fractions.			
simplify expressions written with a fraction bar.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

4.4 Add and Subtract Fractions with Common Denominators

Learning Objectives

By the end of this section, you will be able to:

- Model fraction addition
- Add fractions with a common denominator
- Model fraction subtraction
- Subtract fractions with a common denominator

☑ **BE PREPARED** 4.9 Before you get started, take this readiness quiz.

Simplify: $2x+9+3x-4$.
If you missed this problem, review Example 2.22.

☑ **BE PREPARED** 4.10 Draw a model of the fraction $\frac{3}{4}$.

If you missed this problem, review Example 4.2.

☑ **BE PREPARED** 4.11 Simplify: $\frac{3+2}{6}$.

If you missed this problem, review Example 4.48.

Model Fraction Addition

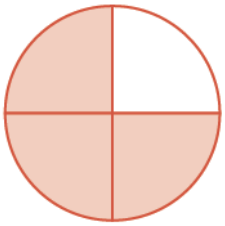
How many quarters are pictured? One quarter plus 2 quarters equals 3 quarters.



Remember, quarters are really fractions of a dollar. Quarters are another way to say fourths. So the picture of the coins shows that

$$\underset{\text{one quarter}}{\frac{1}{4}} + \underset{\text{two quarters}}{\frac{2}{4}} = \underset{\text{three quarters}}{\frac{3}{4}}$$

Let's use fraction circles to model the same example, $\frac{1}{4} + \frac{2}{4}$.

Start with one $\frac{1}{4}$ piece.		$\frac{1}{4}$
Add two more $\frac{1}{4}$ pieces.	+ 	$+\frac{2}{4}$
The result is $\frac{3}{4}$.		$\frac{3}{4}$

So again, we see that

$$\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

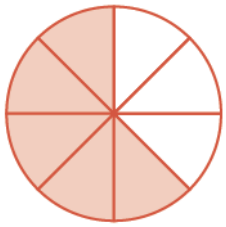
MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Model Fraction Addition" will help you develop a better understanding of adding fractions

EXAMPLE 4.52

Use a model to find the sum $\frac{3}{8} + \frac{2}{8}$.

Solution

Start with three $\frac{1}{8}$ pieces.		$\frac{3}{8}$
Add two $\frac{1}{8}$ pieces.	+ 	$+\frac{2}{8}$
How many $\frac{1}{8}$ pieces are there?		$\frac{5}{8}$

There are five $\frac{1}{8}$ pieces, or five-eighths. The model shows that $\frac{3}{8}+\frac{2}{8}=\frac{5}{8}$.

TRY IT 4.103 Use a model to find each sum. Show a diagram to illustrate your model.

$\frac{1}{8}+\frac{4}{8}$

TRY IT 4.104 Use a model to find each sum. Show a diagram to illustrate your model.

$\frac{1}{6}+\frac{4}{6}$

Add Fractions with a Common Denominator

Example 4.52 shows that to add the same-size pieces—meaning that the fractions have the same denominator—we just add the number of pieces.

Fraction Addition

If a, b, and c are numbers where $c \neq 0$, then

$$\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}$$

To add fractions with a common denominator, add the numerators and place the sum over the common denominator.

EXAMPLE 4.53

Find the sum: $\frac{3}{5}+\frac{1}{5}$.

Solution

	$\frac{3}{5}+\frac{1}{5}$
Add the numerators and place the sum over the common denominator.	$\frac{3+1}{5}$
Simplify.	$\frac{4}{5}$

TRY IT 4.105 Find each sum: $\frac{3}{6}+\frac{2}{6}$.

TRY IT 4.106 Find each sum: $\frac{3}{10}+\frac{7}{10}$.

EXAMPLE 4.54

Find the sum: $\frac{x}{3}+\frac{2}{3}$.

Solution

	$\frac{x}{3}+\frac{2}{3}$
Add the numerators and place the sum over the common denominator.	$\frac{x+2}{3}$

Note that we cannot simplify this fraction any more. Since x and 2 are not like terms, we cannot combine them.

TRY IT 4.107 Find the sum: $\frac{x}{4}+\frac{3}{4}$.

TRY IT 4.108 Find the sum: $\frac{y}{8}+\frac{5}{8}$.

EXAMPLE 4.55

Find the sum: $-\frac{9}{d}+\frac{3}{d}$.

Solution

We will begin by rewriting the first fraction with the negative sign in the numerator.

$-\frac{a}{b}=\frac{-a}{b}$

	$-\frac{9}{d}+\frac{3}{d}$
Rewrite the first fraction with the negative in the numerator.	$\frac{-9}{d}+\frac{3}{d}$
Add the numerators and place the sum over the common denominator.	$\frac{-9+3}{d}$

Simplify the numerator.	$\frac{-6}{d}$
Rewrite with negative sign in front of the fraction.	$-\frac{6}{d}$

TRY IT 4.109 Find the sum: $-\frac{7}{d}+\frac{8}{d}$.

TRY IT 4.110 Find the sum: $-\frac{6}{m}+\frac{9}{m}$.

EXAMPLE 4.56

Find the sum: $\frac{2n}{11}+\frac{5n}{11}$.

Solution

	$\frac{2n}{11}+\frac{5n}{11}$
Add the numerators and place the sum over the common denominator.	$\frac{2n+5n}{11}$
Combine like terms.	$\frac{7n}{11}$

TRY IT 4.111 Find the sum: $\frac{3p}{8}+\frac{6p}{8}$.

TRY IT 4.112 Find the sum: $\frac{2q}{5}+\frac{7q}{5}$.

EXAMPLE 4.57

Find the sum: $-\frac{3}{12}+\left(-\frac{5}{12}\right)$.

Solution

	$-\frac{3}{12}+\left(-\frac{5}{12}\right)$
Add the numerators and place the sum over the common denominator.	$\frac{-3+(-5)}{12}$
Add.	$\frac{-8}{12}$
Simplify the fraction.	$-\frac{2}{3}$

TRY IT 4.113 Find each sum: $-\frac{4}{15}+\left(-\frac{6}{15}\right)$.

TRY IT 4.114 Find each sum: $-\frac{5}{21}+\left(-\frac{9}{21}\right)$.

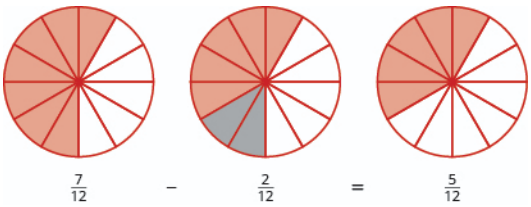
Model Fraction Subtraction

Subtracting two fractions with common denominators is much like adding fractions. Think of a pizza that was cut into 12 slices. Suppose five pieces are eaten for dinner. This means that, after dinner, there are seven pieces (or $\frac{7}{12}$ of the pizza) left in the box. If Leonardo eats 2 of these remaining pieces (or $\frac{2}{12}$ of the pizza), how much is left? There would be 5 pieces left (or $\frac{5}{12}$ of the pizza).

$$\frac{7}{12} - \frac{2}{12} = \frac{5}{12}$$

Let's use fraction circles to model the same example, $\frac{7}{12} - \frac{2}{12}$.

Start with seven $\frac{1}{12}$ pieces. Take away two $\frac{1}{12}$ pieces. How many twelfths are left?



Again, we have five twelfths, $\frac{5}{12}$.

MANIPULATIVE MATHEMATICS

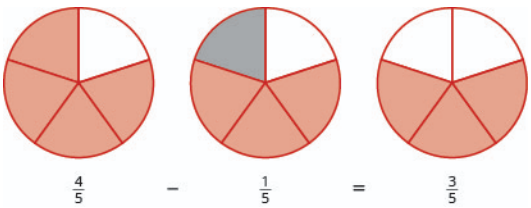
Doing the Manipulative Mathematics activity "Model Fraction Subtraction" will help you develop a better understanding of subtracting fractions.

EXAMPLE 4.58

Use fraction circles to find the difference: $\frac{4}{5} - \frac{1}{5}$.

Solution

Start with four $\frac{1}{5}$ pieces. Take away one $\frac{1}{5}$ piece. Count how many fifths are left. There are three $\frac{1}{5}$ pieces left.



TRY IT 4.115 Use a model to find each difference. Show a diagram to illustrate your model.

$\frac{7}{8} - \frac{4}{8}$

TRY IT 4.116 Use a model to find each difference. Show a diagram to illustrate your model.

$\frac{5}{6} - \frac{4}{6}$

Subtract Fractions with a Common Denominator

We subtract fractions with a common denominator in much the same way as we add fractions with a common denominator.

Fraction Subtraction

If a, b, and c are numbers where $c \neq 0$, then

$$\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$$

To subtract fractions with a common denominator, we subtract the numerators and place the difference over the common denominator.

EXAMPLE 4.59

Find the difference: $\frac{23}{24} - \frac{14}{24}$.

Solution

	$\frac{23}{24} - \frac{14}{24}$
Subtract the numerators and place the difference over the common denominator.	$\frac{23-14}{24}$
Simplify the numerator.	$\frac{9}{24}$
Simplify the fraction by removing common factors.	$\frac{3}{8}$

TRY IT 4.117 Find the difference: $\frac{19}{28} - \frac{7}{28}$.

TRY IT 4.118 Find the difference: $\frac{27}{32} - \frac{11}{32}$.

EXAMPLE 4.60

Find the difference: $\frac{y}{6} - \frac{1}{6}$.

Solution

	$\frac{y}{6} - \frac{1}{6}$
Subtract the numerators and place the difference over the common denominator.	$\frac{y-1}{6}$

The fraction is simplified because we cannot combine the terms in the numerator.

TRY IT 4.119 Find the difference: $\frac{x}{7} - \frac{2}{7}$.

TRY IT 4.120 Find the difference: $\frac{y}{14} - \frac{13}{14}$.

EXAMPLE 4.61

Find the difference: $-\frac{10}{x} - \frac{4}{x}$.

Solution

Remember, the fraction $-\frac{10}{x}$ can be written as $\frac{-10}{x}$.

	$-\frac{10}{x}-\frac{4}{x}$
Subtract the numerators.	$\frac{-10-4}{x}$
Simplify.	$\frac{-14}{x}$
Rewrite with the negative sign in front of the fraction.	$-\frac{14}{x}$

TRY IT 4.121 Find the difference: $-\frac{9}{x}-\frac{7}{x}$.

TRY IT 4.122 Find the difference: $-\frac{17}{a}-\frac{5}{a}$.

Now lets do an example that involves both addition and subtraction.

EXAMPLE 4.62

Simplify: $\frac{3}{8}+\left(-\frac{5}{8}\right)-\frac{1}{8}$.

Solution

	$\frac{3}{8}+\left(-\frac{5}{8}\right)-\frac{1}{8}$
Combine the numerators over the common denominator.	$\frac{3+(-5)-1}{8}$
Simplify the numerator, working left to right.	$\frac{-2-1}{8}$
Subtract the terms in the numerator.	$\frac{-3}{8}$
Rewrite with the negative sign in front of the fraction.	$-\frac{3}{8}$

TRY IT 4.123 Simplify: $\frac{2}{5}+\left(-\frac{4}{5}\right)-\frac{3}{5}$.

TRY IT 4.124 Simplify: $\frac{5}{9}+\left(-\frac{4}{9}\right)-\frac{7}{9}$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Adding Fractions With Pattern Blocks (https://www.openstax.org/l/24AddFraction)

Adding Fractions With Like Denominators (https://www.openstax.org/l/24AddLikeDenom)

Subtracting Fractions With Like Denominators (https://www.openstax.org/l/24SubtrLikeDeno)

SECTION 4.4 EXERCISES

Practice Makes Perfect

Model Fraction Addition

In the following exercises, use a model to add the fractions. Show a diagram to illustrate your model.

254. $\frac{2}{5}+\frac{1}{5}$ **255.** $\frac{3}{10}+\frac{4}{10}$ **256.** $\frac{1}{6}+\frac{3}{6}$

257. $\frac{3}{8}+\frac{3}{8}$

Add Fractions with a Common Denominator

In the following exercises, find each sum.

258. $\frac{4}{9}+\frac{1}{9}$ **259.** $\frac{2}{9}+\frac{5}{9}$ **260.** $\frac{6}{13}+\frac{7}{13}$

261. $\frac{9}{15}+\frac{7}{15}$ **262.** $\frac{x}{4}+\frac{3}{4}$ **263.** $\frac{y}{3}+\frac{2}{3}$

264. $\frac{7}{p}+\frac{9}{p}$ **265.** $\frac{8}{q}+\frac{6}{q}$ **266.** $\frac{8b}{9}+\frac{3b}{9}$

267. $\frac{5a}{7}+\frac{4a}{7}$ **268.** $\frac{-12y}{8}+\frac{3y}{8}$ **269.** $\frac{-11x}{5}+\frac{7x}{5}$

270. $-\frac{1}{8}+\left(-\frac{3}{8}\right)$ **271.** $-\frac{1}{8}+\left(-\frac{5}{8}\right)$ **272.** $-\frac{3}{16}+\left(-\frac{7}{16}\right)$

273. $-\frac{5}{16}+\left(-\frac{9}{16}\right)$ **274.** $-\frac{8}{17}+\frac{15}{17}$ **275.** $-\frac{9}{19}+\frac{17}{19}$

276. $\frac{6}{13}+\left(-\frac{10}{13}\right)+\left(-\frac{12}{13}\right)$ **277.** $\frac{5}{12}+\left(-\frac{7}{12}\right)+\left(-\frac{11}{12}\right)$

Model Fraction Subtraction

In the following exercises, use a model to subtract the fractions. Show a diagram to illustrate your model.

278. $\frac{5}{8}-\frac{2}{8}$ **279.** $\frac{5}{6}-\frac{2}{6}$

Subtract Fractions with a Common Denominator

In the following exercises, find the difference.

280. $\frac{4}{5}-\frac{1}{5}$ **281.** $\frac{4}{5}-\frac{3}{5}$ **282.** $\frac{11}{15}-\frac{7}{15}$

283. $\frac{9}{13}-\frac{4}{13}$ **284.** $\frac{11}{12}-\frac{5}{12}$ **285.** $\frac{7}{12}-\frac{5}{12}$

286. $\frac{4}{21}-\frac{19}{21}$ **287.** $-\frac{8}{9}-\frac{16}{9}$ **288.** $\frac{y}{17}-\frac{9}{17}$

289. $\frac{x}{19}-\frac{8}{19}$ **290.** $\frac{5y}{8}-\frac{7}{8}$ **291.** $\frac{11z}{13}-\frac{8}{13}$

292. $-\frac{8}{d}-\frac{3}{d}$ **293.** $-\frac{7}{c}-\frac{7}{c}$ **294.** $-\frac{23}{u}-\frac{15}{u}$

295. $-\frac{29}{v}-\frac{26}{v}$ **296.** $\frac{6c}{7}-\frac{5c}{7}$ **297.** $\frac{12d}{11}-\frac{9d}{11}$

298. $\frac{-4r}{13}-\frac{5r}{13}$ **299.** $\frac{-7s}{3}-\frac{7s}{3}$ **300.** $-\frac{3}{5}-\left(-\frac{4}{5}\right)$

301. $-\frac{3}{7}-\left(-\frac{5}{7}\right)$ **302.** $-\frac{7}{9}-\left(-\frac{5}{9}\right)$ **303.** $-\frac{8}{11}-\left(-\frac{5}{11}\right)$

Mixed Practice

In the following exercises, perform the indicated operation and write your answers in simplified form.

304. $-\frac{5}{18}\cdot\frac{9}{10}$ **305.** $-\frac{3}{14}\cdot\frac{7}{12}$ **306.** $\frac{n}{5}-\frac{4}{5}$

307. $\frac{6}{11}-\frac{s}{11}$ **308.** $-\frac{7}{24}+\frac{2}{24}$ **309.** $-\frac{5}{18}+\frac{1}{18}$